



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-01 Tool Setup at First Use

Windows/WSL-compatible module tool guide

Artifact	C3_019_C3_01_TOOL_SETUP_FIRST_USE_RC1
Status	2026 - LEARNER EDITION
Teaching standard	M01 beginner-first teacher loop. Videos are visual companions; the book remains sufficient fallback.
C2B boundary	Bridge foundations may be assumed, but C3 engineering behavior is demonstrated, broken, repaired, transferred and validated.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

Course rule: preserve what arrived -> validate interpretation -> quarantine ambiguity -> publish durably -> commit state last -> prove replay safety. A green job is never sufficient evidence by itself.



Use the same C3 project environment from C3-00

Do not create random environments per lesson. Open your C3 project in the Linux-native WSL filesystem if you are using WSL, activate the C3 venv, and install only the dependency first required by the lesson.

L04 - PYARROW

```
python -c "import pyarrow; print(pyarrow.__version__)"
# only if ModuleNotFoundError:
python -m pip install pyarrow
python -c "import pyarrow; print(pyarrow.__version__)"
```

Expected: a version prints. If installation/network is blocked, save the exact error and continue only with conceptual/book work; do not fabricate runtime evidence.

L05 - REQUESTS

```
python -c "import requests; print(requests.__version__)"
# only if missing:
python -m pip install requests
python -c "import requests; print(requests.__version__)"
```

Expected: version prints. The practice mock API stays on localhost; no real credential is required.

L08 - SQLITE3

```
python - <<'PY'
import sqlite3
print(sqlite3.sqlite_version)
PY
```

sqlite3 ships with normal CPython; install nothing separately for this phase.

L14 - PYTEST

```
pytest --version
```

This should already work from C3-00. If it does not, repair the existing environment rather than silently creating a new one.



Troubleshooting ladder

1. AM I IN THE RIGHT SHELL?

On Windows, use your established WSL/terminal path from C3-00. Run ``pwd``, ``python --version``, and inspect the project path.

2. IS THE VENV ACTIVE?

Check ``which python`` on Linux/WSL or ``where python`` in Windows PowerShell. The path should point into the C3 environment when activated.

3. DID PIP INSTALL INTO THE SAME PYTHON?

Prefer ``python -m pip ...`` so pip belongs to the Python you just verified.

4. SAVE THE REAL ERROR

Copy the command + complete error into evidence. "It did not work" is not enough to diagnose.

5. NEVER DOWNGRADE THE LESSON TO A FAKE SUCCESS

If PyArrow or another runtime cannot execute on your machine, mark runtime pending, continue learning from the book, and return when the environment is repaired. Honest blockers preserve trustworthy progress.